# THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

## CLASS X — MATHEMATICS (Standard) · MID-TERM PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

| Class: 10 | Subject: Mathematics | Set 03 | Time: 3 Hours | Max. Marks: 80 |
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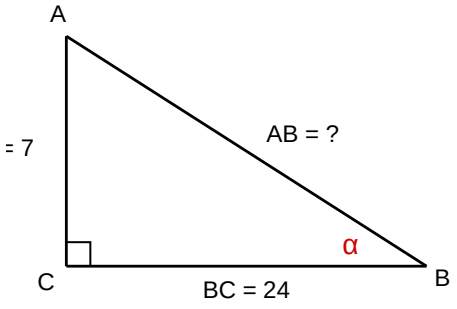
**General Instructions:**

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

### SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

*Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.*

**1.** The value of tan α from the given figure is **1**




(a) $\frac{24}{7}$ (b) $\frac{7}{25}$ (c) $\frac{7}{24}$ (d) $\frac{24}{25}$

**2.** ΔABC ~ ΔPQR. The perimeters of the two triangles are 30 cm and 45 cm. If AB = 12 cm, then PQ is **1**

(a) 8 cm (b) 18 cm (c) 16 cm (d) 27 cm

**3.** The HCF and LCM of 8, 9 and 25 are respectively **1**

(a) 1, 200 (b) 3, 1800 (c) 1, 900 (d) 1, 1800

**4.** Two cousins were both born in February 2024. The probability that they have the same birthday is **1**

(a) $\frac{1}{29}$ (b) $\frac{1}{28}$ (c) $\frac{1}{366}$ (d) $\frac{2}{29}$

**5.** A point lies on the *y*-axis at a distance of 6 units from the origin. Which of the following can be its coordinates? **1**

(a) (6, 0) (b) (−6, 6) (c) (0, −6) (d) (6, 6)

**6.** The value of $(2 + \sqrt{3})^2 + (2 - \sqrt{3})^2$ is **1**

(a) 7 (b) $8\sqrt{3}$ (c) $4 + 7\sqrt{3}$ (d) 14

**7.** The number 7 × 11 × 13 + 13 is **1**

(a) a composite number (b) a prime number
(c) neither prime nor composite (d) an irrational number

**8.** Rohit reads 12 pages on the first day and 4 pages more each day than on the previous day. The number of pages he reads on the 10th day is **1**

(a) 44 (b) 48 (c) 52 (d) 40

**9.** The system $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions. Then $a - b$ equals **1**

(a) 6 (b) 5 (c) 1 (d) 4

**10.** Which of the following is NOT possible for an acute angle θ? **1**

(a) $\sin\theta = \frac{4}{3}$ (b) $\tan\theta = 5$ (c) $\sec\theta = 1.2$ (d) $\cos\theta = 0.8$

**11.** The point P($x$, $y$) divides the line segment joining A(−2, −2) and B(2, −4) in the ratio 3 : 7 internally. Then P is **1**

(a) $(\frac{4}{5}, \frac{13}{5})$ (b) $(\frac{-4}{5}, \frac{13}{5})$ (c) $(\frac{-4}{5}, \frac{-13}{5})$ (d) $(\frac{-8}{5}, \frac{-6}{5})$

**12.** A 2-digit number is selected at random. The probability that it is a multiple of 3 and 5 but not of 2 is **1**

(a) $\frac{1}{15}$ (b) $\frac{1}{30}$ (c) $\frac{1}{18}$ (d) $\frac{1}{10}$

**13.** The solution of $2x + 3y = 12$ and $x - 3y = -3$ is **1**

(a) x = 2, y = 3 (b) x = 6, y = 0 (c) x = 3, y = 2 (d) x = 1, y = 3

**14.** If $x$ and $y$ are distinct primes, the HCF of $x^4y$ and $x^2y^3$ is **1**

(a) $x^2y$ (b) $x^4y^3$ (c) $x^2y^3$ (d) $xy$

**15.** Two dice are thrown simultaneously. The probability that the product of the two numbers is 12 is **1**

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{5}{36}$ (d) $\frac{1}{9}$

**16.** If $\operatorname{cosec}\theta = \frac{13}{12}$, then $\cot\theta$ equals **1**

(a) $\frac{12}{5}$ (b) $\frac{5}{12}$ (c) $\frac{5}{13}$ (d) $\frac{13}{5}$

**17.** The pair of equations $3x + 2y = 7$ and $6x - y = 9$ represents two lines which are **1**

(a) parallel (b) coincident

(c) intersecting at exactly two points (d) intersecting at exactly one point

**18.** The common difference of the AP √2, √8, √18, … is **1**

(a) 2√2 (b) 3√2 (c) √2 (d) 4√2

**19.** **Assertion (A):** The points (2, 3), (4, 5) and (6, 7) are collinear. **1**
**Reason (R):** If PQ + QR = PR for three points P, Q and R, then the points are collinear.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**20.** **Assertion (A):** The probability of an event always lies between 0 and 1, both inclusive. **1**
**Reason (R):** For any event E, P(E) + P(not E) = 1.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

## SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

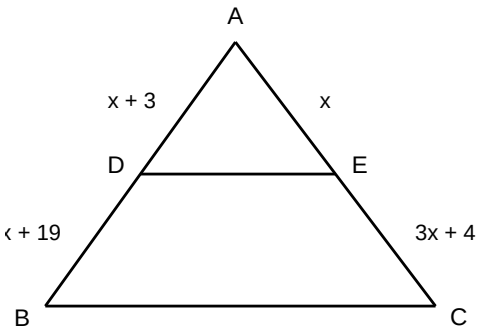
**21.** Find the HCF and LCM of 255 and 867 by the prime factorisation method. **2**

**O R**

Find the smallest number which leaves remainder 2 when divided by 8, 12 and 15.

**22.** A flagpole 18 m high casts a shadow 9 m long. At the same time a building 30 m high casts a shadow of what length? **2**

**23.** In ΔABC, DE ∥ BC with AD = $x$ + 3, DB = 3$x$ + 19, AE = $x$ and EC = 3$x$ + 4. Find $x$. **2**




**O R**

The areas of two similar triangles are 36 cm² and 100 cm². If a side of the first triangle is 9 cm, find the corresponding side of the second.

**24.** A kite is flying at an angle of elevation of 30° with the ground and the string attached to it is taut and 80 m long. Find the height of the kite above the ground. **2**

**25.** The LCM of two numbers is 1200 and their HCF is 40. If one of the numbers is 240, find the other number. **2**

## SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

**26.** Prove that 7√5 is irrational, given that √5 is irrational. **3**

**27.** Solve the pair of equations: $\frac{2}{x} + \frac{3}{y} = 13$; $\frac{5}{x} - \frac{4}{y} = -2$. **3**

**28.** Which term of the AP 3, 8, 13, … is 78? Also find the sum of the terms up to that term. **3**

**O R**

The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of the first $n$ terms.

**29.** Find the coordinates of the point P on the line segment joining A(−6, 10) and B(3, −8) such that P divides AB in the ratio 2 : 7. **3**

**O R**

Show that A(0, −1), B(−2, 3), C(6, 7) and D(8, 3) are the vertices of a rectangle.

**30.** If $\tan\theta = \frac{3}{4}$, find the value of $\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$. **3**

**31.** A box contains 90 discs numbered 1 to 90. One disc is drawn at random. Find the probability that it bears (a) a two-digit number (b) a perfect square (c) a number divisible by 5. **3**

## SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

**32.** State and prove the converse of the Basic Proportionality Theorem. **5**

**33.** Meena withdrew ₹ 2000 from a bank in ₹ 50 and ₹ 100 notes and received 25 notes in all. How many notes of each kind did she receive? **5**

**O R**

Solve graphically: 3$x$ + $y$ = 9 and $x$ − $y$ = −1. Find the points where the lines meet the $y$-axis.

**34.** From a point on a bridge across a river, the angles of depression of the two banks, on opposite sides of the river, are 30° and 45°. If the bridge is 3 m above the water, find the width of the river. (Take $\sqrt{3} = 1.732$) **5**

**35.** Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$. **5**

**OR**

Prove that $\frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\sin\theta} = \frac{1}{\sin\theta} - \frac{1}{\operatorname{cosec}\theta + \cot\theta}$.

## SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

**36.** A pottery workshop makes 12 pots on the first day. It increases production by 3 pots every day. The number of pots made on successive days forms an AP.

(i) State the first term and the common difference of the AP. 1

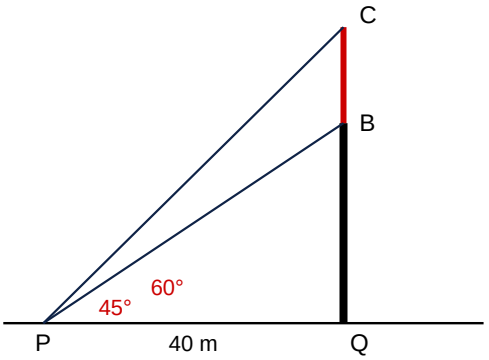
(ii) Find the number of pots made on the 10th day. 1

(iii) On which day will the workshop make 60 pots? 2

**OR**

(iii) Find the total number of pots made in the first 12 days.

**37.** Students of Class X measure the height of their school building. They stand at a point P on level ground 40 m from the foot Q of the building. The angle of elevation of the top B of the building is 45°, and the angle of elevation of the top C of the flagpole fixed on the building is 60°. (Take $\sqrt{3} = 1.732$)




(i) Find the height of the building BQ. 1

(ii) Find the height of the point C above the ground. 1

(iii) Find the length of the flagpole BC. 2

**OR**

(iii) Find the distance of the point P from the top C of the flagpole.

**38.** In a lucky-draw box there are 30 cards bearing the numbers 1 to 30. A card is drawn at random.

(i) Find the probability that the number is a multiple of 6. 1

(ii) Find the probability that the number is a perfect cube. 1

(iii) Find the probability that the number is a prime or a multiple of 10. 2

**OR**

(iii) The card numbered 15 is drawn and not replaced. Find the probability that the next card drawn is a multiple of 5.